Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

MSE Examination

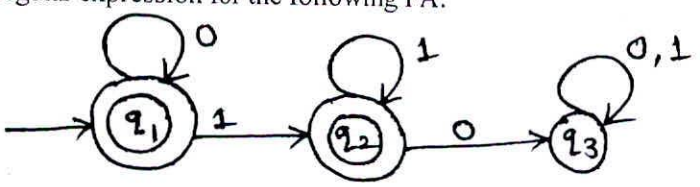
Max. Marks: 30 | October 2023 | Duration: 1hr.

Class: T.E. (V) Computer/CSE-AIML/CSE-DS | Course Code: CS301/AI301/DS301

Name of the Course: Theory of Computation

Instructions:

(1) All questions are compulsory
(2) Partial credit will be given, so show your work where appropriate.
(3) Try to be precise in your answers in order to maximize your points.
(4) Note: DFA = Deterministic Finite Automata, NFA = Nondeterministic Finite Automata, PDA = Push-Down Automata, CFG=Context Free Grammar.
(5) Let w^R denote the reverse of a word w. That is, if w =a1a2 · · · an, then w^R = an · · · a1.

Q. No.		Max Marks	CO
Q 1. (a)	What is the regular expression for the following language L = {w ∈ {a, b}* | w has no two consecutive a's or b's and has at least one a}?	3	1
(b)	How many strings of length at most 3 is contained in the language described by the following regular expression, (x + y)* y (a + ab)*? Specify the language.	3	1
(c)	If L = {0, 10, 11}, which of the following is in L*? Justify your answer. A. 01001. B. 100011. C. 11111. D. 0010111.	2	1
Q2.	Find the regular expression for the following FA. 	3	1
Q. 3	Design a DFA over alphabet Σ = {0,1}, which accepts the set of strings that either start with 01 or end with 01 such that L = {0101, 011101, 01001, 01101 ...}	4	1
Q. 4	State True or False. Give a proof/counterexample supporting your final answer.		
(a)	Deterministic and non-deterministic PDA's have equivalent expressive power.	3	4
(b)	An NFA can recognize any language that a PDA can recognize.	3	4

(c)	The class of context-free languages (CFLs) is closed under union. In particular, if L1, L2 are CFLs, then so is L1 ∪ L2.	3	2
Q. 5	Provide a CFG that generates the language 00*1*	2	3
Q. 6	For the following language L, state whether it is context-free or not. Justify your answers. $L = \{xyx^Ry^R \mid x, y \in \{a, b\}^*\}$	4	2